

KATOOMBA, Australia

WMO: 947440

Lat: 33.712S

Lon: 150.309E

Elev: 1015

StdP: 89.71

Time Zone: 10.00 (AUS)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.0	0.9	-6.1	2.5	7.5	-4.2	3.0	7.2	9.3	2.8	9.0	3.6	1.8	270	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	29.2	17.5	27.2	16.9	25.2	16.5	19.8	24.9	18.8	23.9	18.0	22.6	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	14.9	21.6	17.3	14.0	20.2	16.5	13.3	19.4	61.5	25.0	57.9	24.0	54.9	22.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.2	5.5	-1.6	32.5	0.6	2.2	-2.0	34.2	-2.4	35.5	-2.7	36.7	-3.2	38.4		
(5)			-2.5	21.9	0.5	1.0	-2.9	22.6	-3.1	23.2	-3.4	23.8	-3.7	24.5		

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	17.5	17.4	15.3	12.7	9.5	7.0	5.9	6.9	10.1	12.2	14.0	16.4	(6)
(7)		DBStd	5.11	3.40	3.60	3.08	2.89	2.55	2.39	1.95	2.69	3.35	3.49	4.45	3.62	(7)
(8)		HDD10.0	441	0	1	1	8	39	94	128	102	40	17	11	1	(8)
(9)		HDD18.3	2405	57	56	103	169	273	341	387	353	248	193	146	81	(9)
(10)		CDD10.0	1187	232	207	166	91	24	3	1	7	42	84	129	201	(10)
(11)		CDD18.3	109	31	29	10	1	0	0	0	0	2	14	22		(11)
(12)		CDH23.3	807	244	190	62	3	0	0	0	0	12	110	185		(12)
(13)		CDH26.7	201	71	44	10	0	0	0	0	0	1	33	43		(13)
(14)	Wind	WSAvg	1.7	1.4	1.2	1.3	1.2	1.7	2.2	2.1	2.4	2.2	1.9	1.6	1.4	(14)
(15)	Precipitation	PrecAvg	1390	166	176	177	118	110	100	60	88	68	104	120	117	(15)
(16)		PrecMax	2154	493	545	603	598	320	445	266	581	211	420	430	326	(16)
(17)		PrecMin	749	14	20	13	4	4	1	0	1	5	7	11	3	(17)
(18)		PrecStd	390	110	119	127	119	90	102	56	123	51	91	91	88	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	30.8	28.1	23.5	18.0	15.1	13.2	18.5	21.6	25.2	30.3	30.1	(19)
(20)			MCWB	18.7	18.6	17.5	14.5	12.5	9.9	7.8	11.2	12.8	13.2	15.5	17.2	(20)
(21)		2%	DB	29.1	28.0	24.9	21.0	16.1	13.5	11.5	15.4	19.4	22.7	26.7	28.0	(21)
(22)			MCWB	17.6	18.5	16.8	14.0	11.3	9.6	7.3	9.3	11.5	13.0	15.5	16.5	(22)
(23)		5%	DB	26.3	25.9	22.4	19.2	14.6	12.2	10.2	13.7	17.6	20.6	23.4	25.8	(23)
(24)			MCWB	16.7	18.2	15.6	13.3	10.7	9.1	7.0	8.3	10.7	12.3	14.7	16.0	(24)
(25)		10%	DB	24.1	23.6	20.4	17.5	13.3	11.0	9.1	12.0	15.7	18.6	21.4	23.6	(25)
(26)			MCWB	16.3	17.4	15.2	12.8	10.4	8.4	6.4	7.6	10.0	11.4	14.2	15.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.7	20.1	16.3	14.2	12.0	9.6	12.0	14.0	16.3	17.8	19.8	(27)
(28)			MCDWB	28.0	25.6	23.7	20.1	15.4	13.8	10.7	15.5	19.5	21.6	24.5	25.3	(28)
(29)		2%	WB	19.5	20.1	18.2	15.1	13.0	10.7	8.6	10.3	12.8	14.5	16.9	18.5	(29)
(30)			MCDWB	25.6	24.6	21.9	18.5	14.5	12.3	10.0	14.0	17.2	19.2	22.8	24.0	(30)
(31)		5%	WB	18.4	19.0	17.1	14.2	12.0	9.8	7.8	9.2	11.9	13.6	16.0	17.7	(31)
(32)			MCDWB	23.5	23.9	20.6	17.4	13.4	11.4	9.1	12.1	15.3	18.1	21.9	22.5	(32)
(33)		10%	WB	17.5	18.0	16.0	13.4	11.1	8.9	7.2	8.1	10.9	12.5	15.0	16.7	(33)
(34)			MCDWB	21.7	22.6	19.1	16.4	12.6	10.5	8.5	11.4	14.3	16.7	19.3	21.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	8.2	7.9	7.5	6.0	5.6	5.9	7.4	8.2	9.1	8.7	10.0	(35)
(36)			MCDBR	13.5	12.2	11.5	10.4	8.1	7.4	7.4	9.6	10.1	11.8	12.4	13.2	(36)
(37)			MCWBR	5.6	5.5	5.5	5.3	4.6	4.6	4.6	5.0	5.0	5.4	5.5	5.5	(37)
(38)		5% WB	MCDBR	10.8	10.2	9.6	8.3	5.6	5.9	5.5	8.2	8.5	10.0	10.5	11.1	(38)
(39)			MCWBR	5.6	5.4	5.4	4.8	3.6	4.3	3.9	4.9	4.7	5.7	5.5	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.334	0.313	0.292	0.278	0.267	0.262	0.266	0.292	0.311	0.330	0.330	(40)	
(41)		taud	2.525	2.562	2.597	2.635	2.624	2.672	2.677	2.665	2.568	2.533	2.516	2.534	(41)	
(42)		Ebn at Noon	999	983	969	939	902	891	915	957	976	995	1001	1011	(42)	
(43)		Edh at Noon	111	103	93	79	71	63	66	75	94	105	111	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.53	5.54	4.75	3.87	3.08	2.41	2.81	3.71	4.96	5.78	6.10	6.72	(44)	
(45)		RadStd	0.51	0.54	0.45	0.34	0.21	0.21	0.20	0.25	0.33	0.60	0.59	0.64	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	+0.45	N/A	N/A	+0.76	N/A	N/A	-2	-86	+185	+99

Nomenclature: See separate page